

ECE 635 Homework #8 Answers

1. (a) e^{-jkz} , $n=1$, $R=z$

(b) $\frac{e^{-jkr}}{r}$, $n=3$, $R=r$, take limit as $r \rightarrow 0$

2. $\vec{E}_{\text{scat}} = \frac{-jkE_0 A e^{-jkr}}{4\pi r} \sin\theta \hat{\theta}$, $\vec{H}_{\text{scat}} = \frac{jkE_0 A e^{-jkr}}{4\pi r} \sin\theta \hat{\phi}$

3. $E_z = -\frac{E_0 \omega k_y^2}{k^2} \sin(k_y y) e^{-jk_z z}$

$$E_y = -\frac{E_0 \omega k_y k_z}{k^2} \cos(k_y y) e^{-jk_z z}$$

$$H_x = \frac{E_0}{\mu} k_y \cos(k_y y) e^{-jk_z z}$$

4. $E_x = \pm A_{mn} \frac{\omega k_x k_z}{c} \cos(k_x x) \sin(k_y y) e^{\pm jk_z z}$

$$E_y = \pm A_{mn} \frac{\omega k_y k_z}{k^2} \sin(k_x x) \cos(k_y y) e^{\pm jk_z z}$$

$$E_z = -A_{mn} \frac{j\omega}{k^2} (k^2 - k_z^2) \sin(k_x x) \sin(k_y y) e^{\pm jk_z z}$$

$$H_x = A_{mn} \frac{k_y}{\mu} \sin(k_x x) \cos(k_y y) e^{\pm jk_z z}$$

$$H_y = -A_{mn} \frac{k_x}{\mu} \cos(k_x x) \sin(k_y y) e^{\pm jk_z z}$$

$$H_z = 0$$